

Business Requirement Document

Swiggy Data Analytics

End-to-End Data Pipeline using Microsoft Fabric, SQL, Excel, Python and Power BI

Project Name	Swiggy Data Analytics Project
Document Type	Business Requirement Document (BRD)
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Domain	Food Delivery and Business Intelligence
Technology Used	Microsoft Fabric, Power Bi, Microsoft Azure, MySQL, Python, Excel

1. Executive Summary: -

Swiggy is one of India's largest online food delivery platforms connecting customers, restaurants, and delivery partners across multiple cities. Every day, thousands of orders are placed through the platform, generating large volumes of operational data. This project aims to harness that data and transform it into meaningful, actionable business insights.

This Business Requirements Document (BRD) outlines the scope, objectives, data architecture, technical requirements, and deliverables for building an end-to-end data analytics solution. The solution will integrate data cleaning, dimensional modeling, SQL analysis, and interactive dashboard development using a modern Microsoft cloud technology stack.

2. Business Background: -

A. Business Scenario: -

Swiggy processes thousands of food orders daily across multiple cities and restaurants. Each order generates data related to:

- Customers and their ordering preferences
- Restaurants, menus, and dish-level performance
- Order volume, pricing, and revenue metrics
- Customer ratings and satisfaction scores
- Location-based trends

Management requires a centralized analytics platform to monitor business performance, identify trends, evaluate operational efficiency, and support strategic decision-making.

B. Business Problem Statement: -

Raw Swiggy data is stored in flat files and contains duplicates, missing values, and inconsistent formats. Because the data is unstructured, business stakeholders cannot easily answer critical questions such as:

- Which cities and states generate the highest revenue?
- Which restaurants and dishes are most popular?
- How do customer ratings impact order volume and sales?
- What are the monthly and quarterly order trends?
- What is the distribution of customer spending across price ranges?

A robust analytics solution is required to clean, model, and visualize this data to enable confident, data-driven decision-making.

3. Project Objectives:

The objective of this project is to build a complete, end-to-end data analytics pipeline that delivers the following: -

- Store raw swiggy datasets securely in a Lakehouse (raw data storage layer).
- Clean and validate the data using SQL to ensure data quality and consistency.
- Model the transformed data into a Star Schema inside a Data Warehouse.
- Create a Power BI Semantic Model with defined relationships, measures, and KPIs.
- Build interactive Power BI and Excel dashboards for business stakeholders.
- Publish the final reports to Microsoft Fabric for stakeholder access.

A. Key Deliverable: -

- Ingested and pre-processed raw swiggy datasets in the Lakehouse.
- Cleaned and validated data using Python and SQL.
- Designed and implemented Star Schema data warehouse.
- Developed KPIs and analytical measures using SQL and DAX.
- Built interactive dashboards in both Excel and Power BI.
- Published final reports to Microsoft Fabric for stakeholder access.

4. Dataset Information: -

The dataset used in this project is a hybrid dataset created for analytical and learning purposes. It combines publicly available swiggy data with simulated order-level transactions to replicate a real-world food delivery business environment. It is specifically designed to support Star Schema modeling and KPI analysis.

5. Data Cleaning and Validation: -

- Checked for null values in all critical columns
- Identified blank and empty string values
- Standardized date formats
- Detected duplicate rows using GROUP BY and COUNT(*)
- Removed duplicate records using temporary / surrogate keys
- Validated price and rating values
- Ensured data consistency across all columns

6. Step-by-Step Actions Taken from Raw Data to Actionable Insights: -

- Collected and combined source datasets.
- Used Python (Pandas) to generate and transform order-level data.
- Imported cleaned CSV data into MySQL.
- Performed data profiling to understand structure and quality.
- Executed SQL queries to identify nulls, blanks, and duplicates.
- Removed duplicate rows and standardized formats.
- Designed a star schema with dimension and fact tables.
- Loaded unique values into dimension tables.
- Populated the fact table by resolving foreign keys.
- Created KPI queries and analytical measures.
- Connected the model to Excel and Power BI.
- Built dashboards with slicers, charts, and KPIs.
- Published the final report to Microsoft Fabric.
- Delivered insights for business stakeholders.

Dataset Source Files (.CSV): -

File Name	Table Type	Key Columns
Fact_Table_Orders	Fact	Order_id, date_id, location_id, restaurant_id, dish_id, price, rating, rating_count
Dimention_Table_Date	Dimension	date_id, order_date
Dimention_Table_Dishesh	Dimension	dish_id, category, dish_name
Dimention_Table_Locations	Dimension	location_id, state, city, location
Dimention_Table_Restaurant	Dimension	restaurant_id, restaurant_name

7. Technology Stack: -

Layer	Technology	Purpose
Data Storage	Data Lakehouse (Microsoft Fabric)	Raw file storage for CSV datasets
Data Cleaning	MySQL	Data validation, find deduplication, and formatting
Data Processing	Python (Pandas)	Data transformation, and pre-processing
Data Modeling	Data Warehouse (Microsoft Fabric)	Star Schema design and dimensional modeling
Semantic Layer	Power Bi Semantic Model	Table relationships, measures, and KPIs
Visualization	Power Bi and Excel	Interactive dashboards and KPI reports
Cloud Environment	Microsoft Azure	Cloud hosting
Publishing	Microsoft Fabric	Report sharing and stakeholder access

8. Data Architecture and Flow: -

A. Workspace Setup: -

A dedicated Microsoft Fabric workspace named Swiggy Analytics will be created. This workspace will contain:

- Lakehouse - Raw data storage layer
- Data Warehouse - Dimensional model layer
- Data Pipelines - Data transformation
- Power BI Semantic Model - Business logic and KPIs
- Power BI Report - Interactive dashboards

B. Data Pipeline Steps: -

I. Data Lakehouse: -

Raw CSV datasets will be uploaded to the Lakehouse in the following folder structure:

Files → Upload Raw Data Files → Fact_Table_Orders, Dimention_Table_Date, Dimention_Table_Dishesh, Dimention_Table_Locations, Dimention_Table_Restaurant

II. Data Cleaning and Validation (SQL): -

SQL queries will be executed against the raw data to ensure quality the tasks includes: -

- Null Check - Identify missing values across all critical columns such as (State, City, Order_Date, Restaurant_Name, Category, Dish_Name, Price, Rating, Rating_Count).
- Blank / Empty String Check - Detect fields containing empty values that could invalidate analysis.
- Duplicate Detection - Identify duplicate rows using GROUP BY on all business-critical columns.
- Duplicate Removal - Use temporary / Surrogate key to retain one clean copy of each unique order and delete duplicates.
- Date Standardization - Consistent date formats (MM-DD-YYYY).

III. Dimention Modelling (Star Schema): -

The cleaned data will be transformed into a Star Schema inside the Data Warehouse. This structure separates descriptive attributes into dimension tables and concentrates measurable values in a central fact table, enabling faster queries, reduced duplication, and clean aggregations.

Dimension Tables: -

- Dimention_Table_Date
- Dimention_Table_Location
- Dimention_Table_Restaurant
- Dimention_Table_Category
- Dimention_Table_Dishesh

Fact Table: -

- Fact_Table_Orders — Order_id, Date_id, Location_id, Restaurant_id, Dish_id, Price (DECIMAL), Rating (DECIMAL), Rating_Count (INT).

IV. Semantic Model (Power Bi): -

A Power BI Semantic Model will be created of Data Warehouse with the following definitions:-
Table relationships between all dimension tables and the fact table.

- DAX measures for KPIs (Total Orders, Total Revenue, Average Order Value, Average Rating).

9. KPI's Requirements: -

A. Core KPI's Analysis: -

KPI Name	Formula	Business Value
Total Orders	COUNT(Order_id)	Measures total order volume
Total Revenue	SUM(Price)	Tracks overall platform revenue in INR
Average Order Value	AVG(Price)	Indicates customer average spend per order
Average Rating	AVG(Rating)	Reflects overall customer satisfaction

B. Date – Based Analysis: -

- Monthly order trends and revenue
- Quarterly order volume trends
- Year wise growth
- Day-of-week order patterns
- Weekend vs. weekday analysis

C. Location – Based Analysis: -

- Top 5 cities by order volume
- Revenue contribution by state

D. Food – Based Analysis:-

- Top 5 restaurants by order count
- Top categories (Dessert, Main Course, Burgers, Sweet etc.)
- Top ordered dishes
- Dish performance. (Orders + Average Rating)

E. Customer Spending Analysis: -

Distribution of Orders:-

- Under Rs.100
- Rs.100 – Rs.199
- Rs.200 – Rs.299
- Rs.300 – Rs.499
- Above Rs.500+

F. Rating Analysis: -

- Distribution of dish ratings from 1 to 5.

10. Dashboard Requirement: -

The dashboard must provide the following interactive filters:

- State and City
- Restaurant Name
- Food Category
- Dish Name
- Date Range (Year, Quarter, Month)

11. Visual Components: -

- KPI Cards - Total Sales, Average Order Value, Total Orders, Average Rating, Rating Count.
- Line/Area Charts - Monthly and quarterly trend analysis.
- Bar Charts - Top restaurants, top cities.
- Pie / Donut Charts - Veg vs. Non-Veg, category distribution.
- Distribution Analysis - Customer spend buckets and ratings breakdown.
- Map Visual - Geographic revenue distribution by state/city.

12. Expected Business Insights: -

- Revenue-driving cities and states
- Best-performing restaurants
- Seasonal ordering patterns
- Relationship between ratings and order volume
- Customer spending behaviour